

Aditi Roy Adri

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

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Objective

Motivated Computer Science and Engineering student at BRAC University with a strong foundation in algorithms, machine learning, systems programming, and full-stack development. Seeking opportunities to apply technical expertise, contribute to impactful projects, and grow in a collaborative environment.

Education

- BRAC University, Dhaka** 2023 – 2026
Bachelor of Science in Computer Science and Engineering **CGPA: 3.84**
- Holy Cross College, Dhaka** 2021
Higher Secondary Certificate (HSC) **GPA: 5.00**
- Narayanganj High School and College, Narayanganj** 2019
Secondary School Certificate (SSC) **GPA: 5.00**

Technical Skills

- Programming Languages:** Python, C++, C, Java, PHP, Assembly (Emu8086), Bash
- Machine Learning & Computer Vision:** Deep Learning, CNNs, Image Segmentation, Object Detection, Transfer Learning, Model Evaluation, YOLO, U-Net
- Web Technologies:** HTML5, CSS3, JavaScript, React.js
- Databases:** MySQL, PostgreSQL
- Libraries & Frameworks:** PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, Jupyter Notebook
- Graphics & Simulation:** OpenGL (PyOpenGL), Cisco Packet Tracer
- Tools & Platforms:** Git, GitHub, LaTeX, Linux, VS Code
- Soft Skills:** Communication, Critical Thinking, Problem-Solving, Teamwork

Projects

- CARDD – Car Damage Detection with YOLO** Jun 2026
Trained and benchmarked YOLO11-L against YOLOv8-L on the CarDD dataset (4,000 images, 8,740 annotated boxes across 6 damage classes: scratches, dents, cracks, broken lamps, shattered glass, flat tires). YOLO11-L achieved **mAP@50–95 of 0.557**.
Tech: Python, YOLO11, YOLOv8, PyTorch, Object Detection [GitHub](#)
- Pet Image Segmentation & Breed Classification (U-Net)** May 2026
Deep learning project using U-Net and Attention U-Net architectures for binary pet image segmentation and 37-breed classification on the Oxford-IIIT Pet Dataset. Achieved **~88% IoU** with a pretrained ResNet-34 encoder. Implemented data augmentation, combined loss functions, and ablation studies.
Tech: Python, PyTorch, Jupyter Notebook [GitHub](#)
- SaveLife – SOS Emergency Response App** Jul 2026
Emergency response platform with one-tap SOS alerts and GPS geolocation, a live responder dash-

board, and role-based JWT authentication. React/Vite frontend, Node.js/Express API, PostgreSQL, deployed on Render.

Tech: React, Vite, Node.js, Express, PostgreSQL, JWT [GitHub](#)

- **Roaming Sonic – Travel & Tour Management Platform** *Jan 2026*
Full-stack MERN web application for Bangladesh tourism featuring tour packages, hotel bookings, bus tickets, guide services, AI chatbot, and referral rewards system. Live-deployed on Vercel.
Tech: MongoDB, Express.js, React.js, Node.js [GitHub](#) | [Live Demo](#)
- **Metadata Journal System** *Jan 2026*
OS-level project implementing file metadata management and journaling mechanisms for maintaining filesystem consistency.
Tech: C [GitHub](#)

Experience & Activities

- Women & Child Affairs Secretary, Executive Committee — Bangladesh Youngster Social Organization (BYSO), Dhaka District Unit, for the 2026–27 session. Leadership role supporting women’s and children’s welfare initiatives.
- Competitive programming on Codeforces (500+ problems solved in Python).
- Participation in university-level and inter-university programming contests at BRAC University.
- Active volunteer in community and social-welfare activities.
- Vocalist for cultural clubs and events at college and university.

Interests

Machine learning, image processing, computer graphics, competitive programming, full-stack web development, movies and TV series.

References

- **Upal Aditya Oikya, PhD** *Assistant Professor*
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- **Ipshita Bonhi Upoma** *Lecturer*
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